

BURSA, Turkey

WMO: 171160

Lat: 40.2308N

Lon: 29.0133E

Elev: 100

StdP: 100.13

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.1	(c) -1.8	(d) -6.0	(e) 2.3	(f) -0.2	(g) -4.8	(h) 2.6	(i) 0.9	(j) 9.2	(k) 13.7	(l) 8.0	(m) 14.1	(n) 1.3	(o) 60	(p) 0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.4	(c) 34.6	(d) 22.2	(e) 33.2	(f) 22.0	(g) 31.9	(h) 21.6	(i) 24.1	(j) 31.9	(k) 23.3	(l) 30.8	(m) 22.6	(n) 29.9	(o) 3.8	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.5	28.5	20.8	15.7	27.8	20.1	14.9	26.9	72.9	31.7	69.8	30.9	67.1	30.0	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 6.4	(c) 5.5	(e) -6.8	(f) 38.4	(g) 2.5	(h) 1.9	(i) -8.7	(j) 39.7	(k) -10.1	(l) 40.9	(m) -11.6	(n) 41.9	(o) -13.4	(p) 43.3		
(a) 7.5	(b) 6.4	(c) 5.5	(e) -7.1	(f) 25.2	(g) 2.5	(h) 1.6	(i) -8.9	(j) 26.4	(k) -10.4	(l) 27.3	(m) -11.8	(n) 28.3	(o) -13.6	(p) 29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	5.8	7.3	9.3	13.6	18.3	22.6	25.1	25.3	21.2	16.1	11.2	7.0		
	DBStd	7.77	4.18	4.31	4.21	3.87	3.16	2.73	2.05	2.19	2.98	3.53	4.07	4.11		
	HDD10.0	459	142	96	65	11	0	0	0	0	0	2	32	113		
	HDD18.3	1831	388	309	280	147	39	2	0	0	9	87	217	353		
	CDD10.0	2391	12	20	45	119	259	378	469	474	337	191	67	18		
	CDD18.3	721	0	1	1	6	39	131	211	216	96	17	2	1		
	CDH23.3	7283	0	2	15	97	419	1323	2249	2220	811	135	12	0		
	CDH26.7	2982	0	0	2	21	114	524	1014	1002	275	28	1	0		
(14) Wind	WSAvg	2.0	2.1	2.3	2.3	2.0	2.0	2.0	2.3	2.4	1.8	1.5	1.6	2.1		
(15) Precipitation	PrecAvg	702	89	72	70	66	46	37	15	18	40	71	76	111		
	PrecMax	1328	210	179	148	149	128	135	72	84	132	397	142	244		
	PrecMin	448	8	7	5	3	3	2	0	0	0	2	2	7		
	PrecStd	137	52	41	34	36	30	28	16	22	36	59	35	56		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	21.9	25.4	29.1	32.3	36.0	37.8	36.1	34.3	29.6	24.9	20.2		
		MCWB	11.9	13.0	15.0	17.9	20.3	22.6	23.2	23.0	21.3	18.7	15.4	13.5		
	2%	DB	16.8	18.8	21.8	26.0	29.6	33.2	34.6	34.3	31.7	26.8	22.1	17.9		
		MCWB	10.3	11.7	13.2	16.6	19.4	21.5	22.4	22.3	20.5	18.3	14.8	12.4		
	5%	DB	14.9	16.7	19.5	23.5	27.6	31.4	33.0	33.0	29.8	24.5	20.2	15.5		
		MCWB	9.9	10.9	12.4	15.6	18.4	20.9	22.2	22.3	20.3	17.5	13.9	11.0		
	10%	DB	12.4	14.7	17.0	21.2	25.6	29.8	31.6	31.8	28.0	22.3	17.9	13.1		
		MCWB	8.7	10.0	11.2	14.5	17.7	20.3	21.7	21.8	19.4	16.5	13.3	10.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	14.2	16.4	19.0	21.7	24.2	25.1	25.3	23.2	20.9	17.2	14.2		
		MCDB	17.5	19.9	23.9	26.5	29.8	32.6	33.4	33.4	30.5	26.7	22.4	18.4		
	2%	WB	11.0	12.4	13.9	17.4	20.2	22.8	24.0	24.2	22.0	19.2	15.7	13.0		
		MCDB	15.6	17.3	20.0	24.4	27.6	30.4	32.0	31.7	29.0	24.5	20.1	17.2		
	5%	WB	10.0	11.1	12.6	16.0	19.2	21.9	23.2	23.4	21.1	18.2	14.6	11.5		
		MCDB	14.1	15.9	18.3	22.2	25.9	29.3	30.9	30.7	27.5	22.9	18.7	14.8		
	10%	WB	9.0	10.1	11.5	14.8	18.2	21.0	22.6	22.6	20.2	17.2	13.5	10.2		
		MCDB	12.5	14.3	16.5	20.7	24.3	28.1	30.0	29.7	26.2	21.3	17.5	12.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.3	9.7	11.1	11.7	12.2	12.4	12.2	11.8	10.3	9.4	7.1		
		MCDBR	10.5	12.4	14.1	15.5	15.4	15.2	14.9	14.1	14.9	13.7	12.7	9.4		
		MCWBR	6.0	7.1	7.6	8.0	7.2	6.2	5.8	5.4	6.0	6.4	6.8	5.5		
	5% WB	MCDBR	9.7	11.5	12.8	14.2	13.5	13.4	12.8	12.4	13.0	11.6	11.0	7.8		
		MCWBR	5.9	6.8	7.4	7.9	7.0	6.3	5.6	5.4	5.9	6.0	6.4	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.356	0.386	0.428	0.471	0.442	0.423	0.424	0.430	0.418	0.402	0.368	0.357		
	taud		2.350	2.248	2.145	2.049	2.205	2.299	2.309	2.289	2.309	2.332	2.374	2.367		
	Ebn at Noon		789	813	818	807	841	856	851	835	820	789	769	755		
	Edh at Noon		89	112	139	163	142	130	127	127	117	103	86	81		
(44) All-Sky Solar Radiation	RadAvg		1.65	2.37	3.61	4.92	6.19	7.07	7.40	6.45	4.85	3.23	2.17	1.53		
	RadStd		0.23	0.35	0.35	0.55	0.50	0.58	0.26	0.35	0.37	0.28	0.27	0.23		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	-98	N/A	+44

Nomenclature: See separate page